Analysis of massive data sets

http://www.fer.hr/predmet/avsp

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Recommender Systems

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Outline

- Motivation
- Formal Definition
- Content-based Recommendation
- Collaborative Filtering
- Remarks, Practice & Advices

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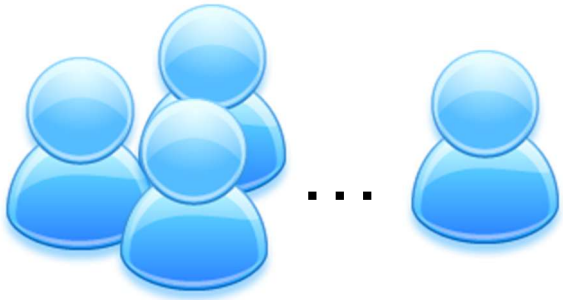
Motivation

- **Example**
  - Music CDs web shop
  - User A
    - Buys Metallica CD
    - Buys Iron Maiden CD
  - User B
    - Does search on Metallica
    - Recommender suggests Iron Maiden from the data collected about User A

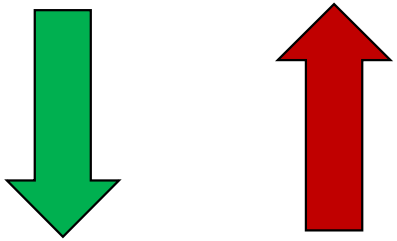
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Motivation

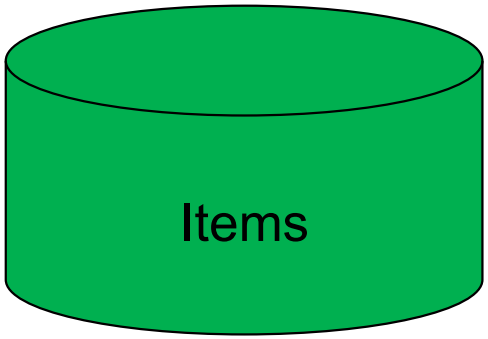
- **Recommender Systems**



Consume



Recommendations

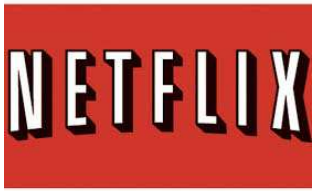


Products, Web sites,
Songs, Videos, News…

Examples

















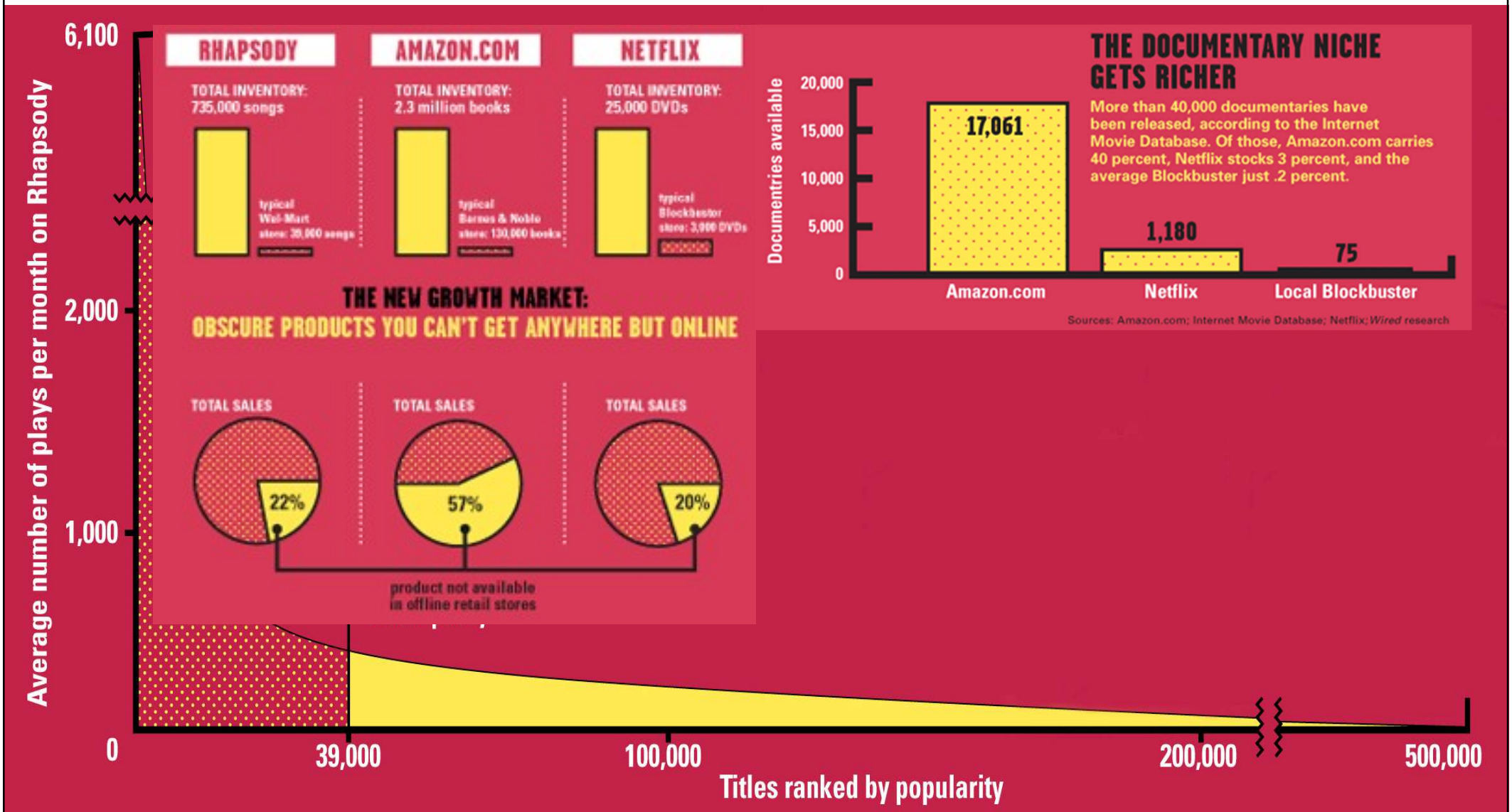
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Motivation

- **From Scarcity to Abundance**
  - Shelf space is a limited resource to traditional retailers
    - Similar: TV advertisement, movie theaters etc.
  - Web enables almost zero-cost dissemination of information about products
    - No issue with limited space
    - Moreover, product information can be personalized
  - The more choice there is, better filter are required
    - Recommendation engines
    - How ***Into Thin Air*** made ***Touching the Void*** a bestseller http://archive.wired.com/wired/archive/12.10/tail.html

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Motivation



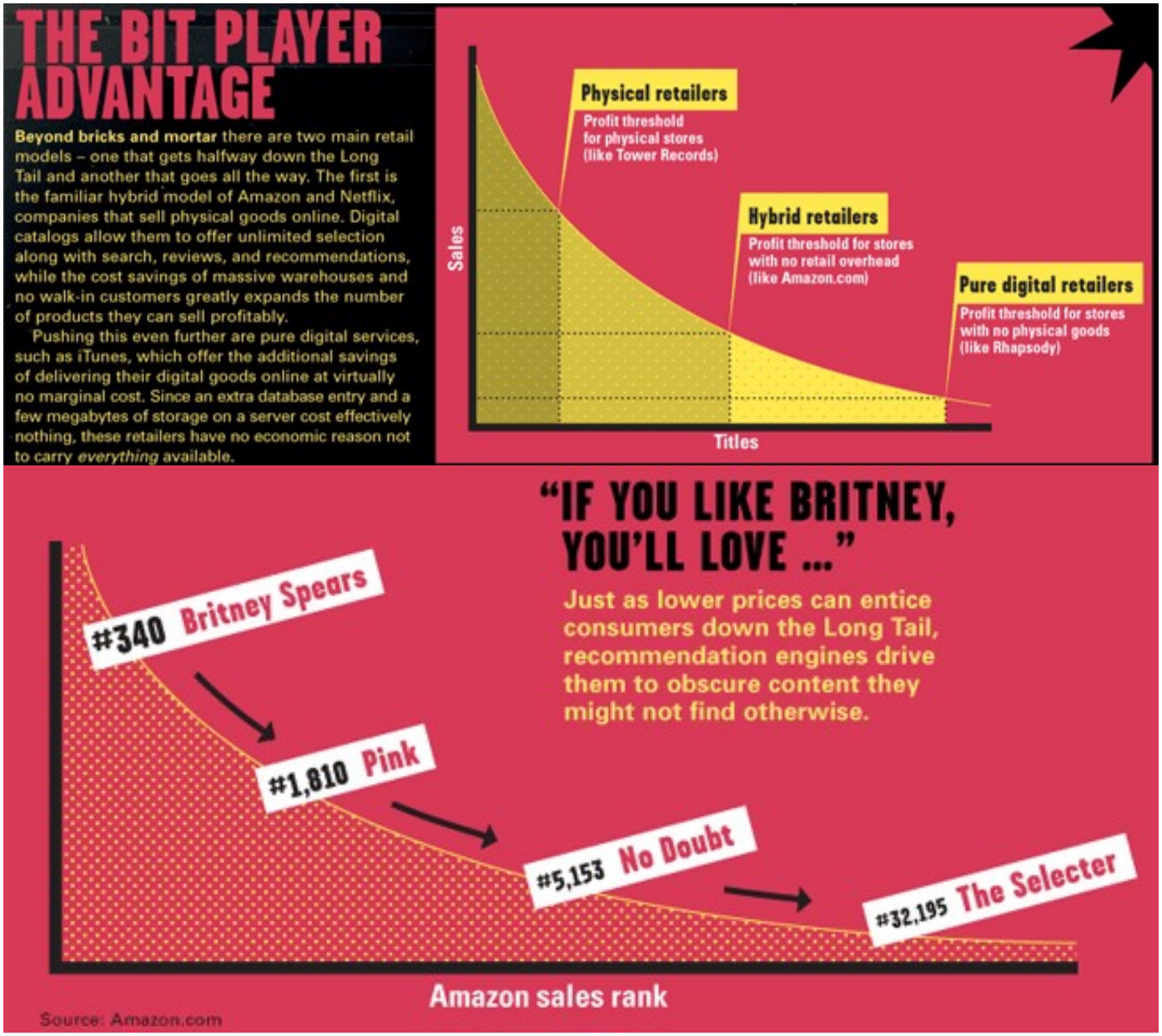
Sources: Erik Brynjolfsson and Jeffrey Hu, MIT, and Michael Smith, Carnegie Mellon; Barnes & Noble; Netflix; RealNetworks

Source: Chris Anderson (2004)

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Motivation

Read & Learn More



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Motivation

- **Types of Recommendations**
  - Editorial and hand selected
    - List of favorites
    - List of “must see” items
  - Simple aggregates
    - Top 10
    - Most Popular
    - Most Recent
  - Personalized recommendations
    - Amazon, EBay, YouTube etc…

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Formal Definition

- **Formal model**
  - X is a set of customers
  - Y is a set of items
  - R is a set of ratings

  - We define a utility function u: $X \times Y \rightarrow R$
  - R is a totally ordered set
    - E.g. 0-5 stars, real number $\in [0, 1]$ etc.

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Formal Definition

- **Utility Matrix**
  - Then, we define a utility matrix which maps users and ratings

	Avatar	LOTR	Matrix	Pirates
Alice				
Bob				
Carol				
David				

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Formal Definition

- **Utility Matrix**
  - Then, we define a utility matrix which maps users and ratings

	Avatar	LOTR	Matrix	Pirates
Alice	1.0		0.2	
Bob		0.5		0.3
Carol	0.2		1.0	
David				0.4

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Formal Definition

- **Main Challenges**
  - Challenge #1: Collecting “known” ratings in matrix
    - How to collect the initial data?
  - Challenge #2: Utilize known ratings to estimate/predict the unknown ratings
    - Focus on high unknown ratings
    - The idea is to rather extract the information what a particular user likes than dislikes
  - Challenge #3: Evaluate prediction methods
    - How to measure accuracy & performance of recommendation methods

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Formal Definition

- **Challenge #1: Collecting Initial Ratings**
  - Explicit
    - Ask people to rate items
    - Not effective in practice
    - Users don’t like to provide feedback explicitly
  - Implicit
    - Learn ratings from users’ actions
      - E.g., purchase implies high rating
    - How can one implicitly detect low ratings?

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Formal Definition

- **Challenge #2: Utilize available ratings to predict missing ones**
  - Data sparsity: Utility matrix U is extremely sparse
    - Ordinary user rates only a very limited subset of items
  - Cold start issue:
    - New users don't have rating history
    - New items don't have any ratings
  - Most effective approaches to overcome the challenges in recommender systems
    - Content based recommendation
    - Collaborative filtering
    - Latent factor analysis

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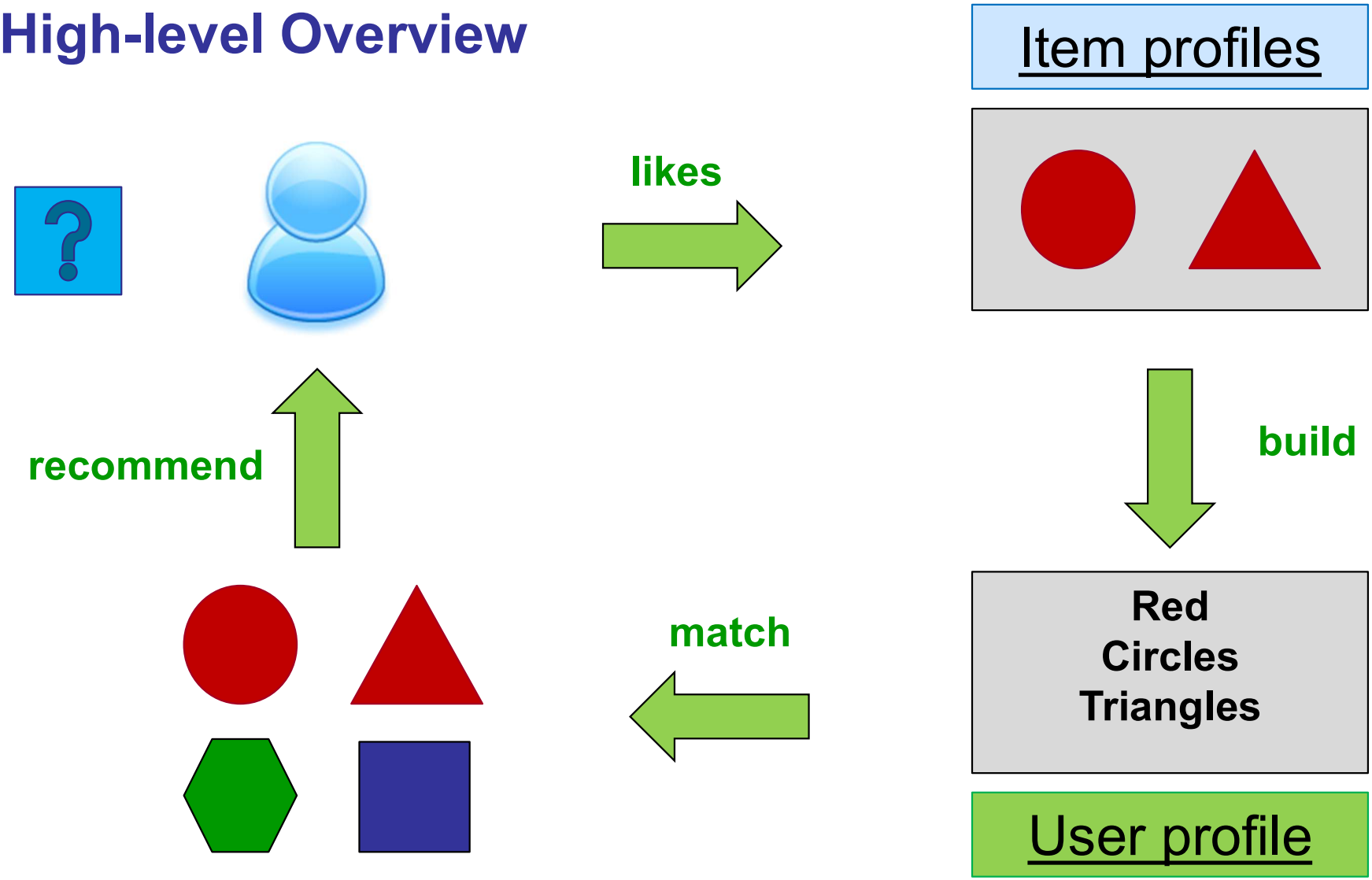
Content-based Recommender Systems

- **Main idea**
  - Recommend user A items that are like previous items highly rated by user A
- **Example**
  - Movie recommendations
    - Recommends movies with same actor(s) director, genre, etc.
  - Websites, blogs, news
    - Recommend other sites with similar content

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Content-based Recommender Systems

- High-level Overview



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Content-based Recommender Systems

- **Item Profiles**
  - For each item, create an item profile
  - Profile is a set (vector) of features
    - Movies: author, title, director, genre…
    - Text: Set of "important" words in document
  - How to pick important features?
    - Usual heuristic from text mining is TF-IDF
      (Term frequency * Inverse Document Frequency)
    - Term … Feature
    - Document … Item

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Content-based Recommender Systems

- **TF-IDF**
  - f_{ij} is a frequency of term (feature) i in a doc (item) j

$$TF_{ij} = \frac{f_{ij}}{\max_k f_{kj}}$$

  - n_i is a number of docs that contain item i
  - N is a total number of docs

$$IDF_i = \log \frac{N}{n_i}$$

  - TF-IDF score $w_{ij} = TF_{ij} \times IDF_i$
  - Document profile:
    - set of words with highest TF-IDF scores along with their scores

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Content-based Recommender Systems

- **User Profiles and Prediction**
  - User profile possibilities
    - Weighted average of rated item profiles
  - Prediction heuristic
    - Estimate $u(i,j)$ for user i and item j:

$$u(i,j) = cosine(i,j) = \frac{i \cdot j}{\|i\| \cdot \|j\|}$$

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Content-based Recommender Systems

- **User Profiles Example**
  - User has rated 5 movies
    - 2 starring actor A, ratings 3 and 5
    - 3 starring actor B, ratings 1, 2 and 4
  - Let's build user profile
    - First, notice that ratings are relative,
      - For one user rating 4 might be a high rating
      - For some other user rating 4 might be an average rating
    - So, first normalize ratings by subtracting the mean which is 3
      - Actor A, 0 and 2, Actor B, -2, -1 and 1
    - Weight of feature A becomes (2 + 0) / 2 = 1
    - Weight of feature B becomes (-2 -1 + 1) / 3 = -2 / 3
    - User profile is A + (-2/3) * B

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Content-based Recommender Systems

- **Pros: Content-based RS**
  - No need for data on other users
    - No cold-start and sparsity issues
  - It can recommend to users with unique tastes
  - It can recommend new & unpopular items
    - No first-rater issue
  - Transparency
    - It can provide transparent explanation about content features that caused an item to be recommended

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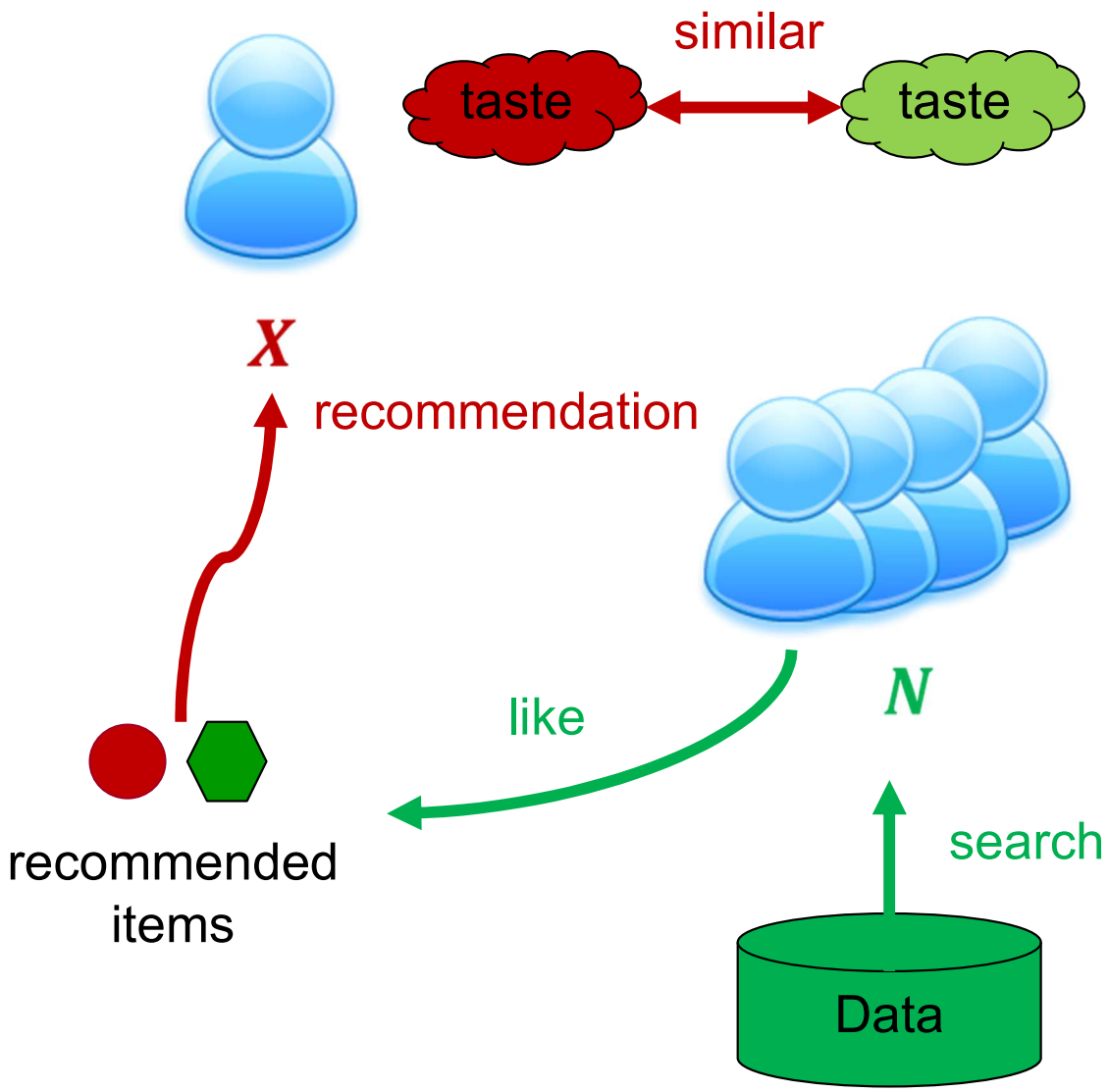
Content-based Recommender Systems

- **Cons: Content-based RS**
  - Finding appropriate features is difficult and requires domain-specific knowledge
    - E.g., movies, images, books etc.
  - It can not recommend for new users
    - How to build user profile for a new user?
  - Overspecialization
    - Recommends only items that match user's content profile
    - People might have multiple interests
    - **It can not utilize ratings from other users**

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Collaborative Filtering

- Imagine a user X
- Find a set N containing users whose ratings are similar to X's ratings
- Estimate X's ratings based on ratings of users in N



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Collaborative Filtering

- **Finding Similar Users**
  - Let vector r_x represent user X's ratings
  - Jaccard similarity measure
    - Issue: It does not consider the values of ratings
  - Cosine similarity measure
    - $sim_cosine(x, y) = cosine(r_x, r_y) = \frac{r_x \cdot r_y}{\|r_x\| \cdot \|r_y\|}$
    - Issue: It threats missing ratings as "negative"
  - Pearson correlation coefficient
    - Let S_{xy} be the set of items rated by both user X and Y

$$sim_pearson(x, y) = \frac{\sum_{s \in S_{xy}} (r_{xs} - \overline{r_x})(r_{ys} - \overline{r_y})}{\sqrt{\sum_{s \in S_{xy}} (r_{xs} - \overline{r_x})^2 \sum_{s \in S_{xy}} (r_{ys} - \overline{r_y})^2}}$$

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Collaborative Filtering

- **Similarity Metric**

	I1	I2	I3	I4	I5	I6	I7
A	4			5	1		
B	5	5	4				
C				2	4	5	

  - Intuitively we want: $sim(A, B) > sim(A, C)$
    - Jaccard similarity produces: 1/5 < 2/4

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Collaborative Filtering

Similarity Metric

	I1	I2	I3	I4	I5	I6	I7
A	4	0	0	5	1	0	0
B	5	5	4	0	0	0	0
C	0	0	0	2	4	5	0

- Intuitively we want: $sim(A, B) > sim(A, C)$
  - Jaccard similarity produces: 1/5 < 2/4
  - Cosine similarity produces: 0.378 > 0.322
    - It threats missing values as negatives

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Collaborative Filtering

Similarity Metric

	I1	I2	I3	I4	I5	I6	I7
A	2/3	0	0	5/3	-7/3	0	0
B	1/3	1/3	-2/3	0	0	0	0
C	0	0	0	-5/3	1/3	4/3	0

- Intuitively we want: $sim(A, B) > sim(A, C)$
  - Jaccard similarity produces: 1/5 < 2/4
  - Cosine similarity produces: 0.378 > 0.322
    - It threats missing values as negatives
  - Pearson similarity produces: 0.092 > - 0.559
    - It subtracts row mean prior to computing cosine

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Collaborative Filtering

- **Rating Predictions**
  - Let vector r_x represent user X's ratings
  - Let N be the set of k most similar users that rated item i
  - Prediction of X's ratings on item i
    - $r_{xi} = \frac{1}{k} \sum_{y \in N} r_{yi}$
      - Threats each similar user contribution equally
    - $r_{xi} = \frac{\sum_{y \in N} sim_{xy} \cdot r_{yi}}{\sum_{y \in N} si_{xy}}$
      - Enables more similar users to contribute with greater weight
    - There are other options

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Collaborative Filtering

- **Item-Item Collaborative Filtering**
  - So far, we considered User-User CF
  - Another perspective: Item-Item
    - For item i, find set of similar items N
    - Estimate rating for item i, based on ratings for similar items
    - Similarity measure and prediction calculations remains the same as in user-user model

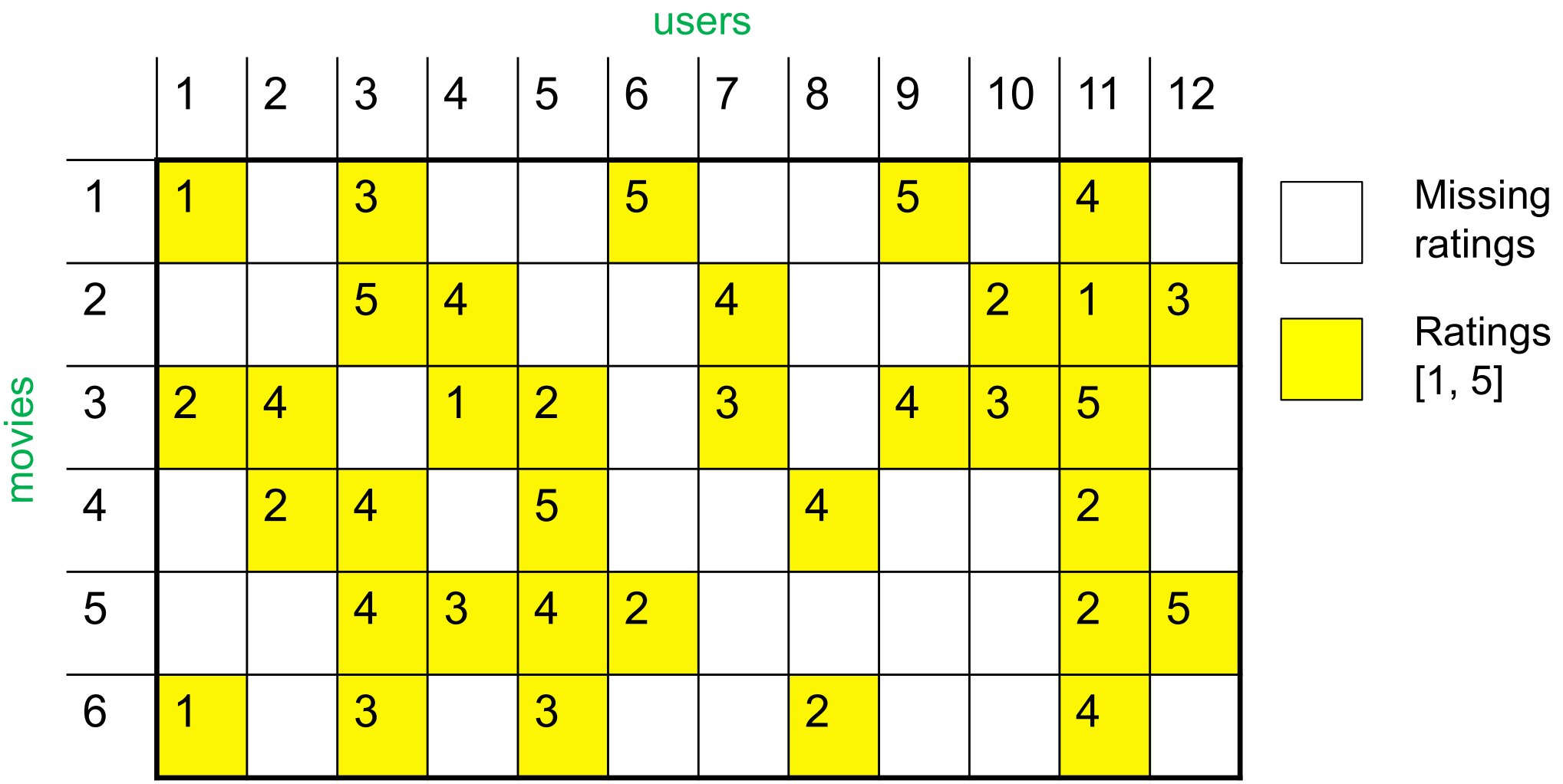
$$r_{xi} = \frac{\sum_{j \in N} sim_{ij} \cdot r_{xj}}{\sum_{j \in N} si \quad ij}$$

sim_{ij} is a similarity of items i and j
r_{xj} is a rating of user x on item j
N is a set of most similar items to i, rated by x

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Collaborative Filtering

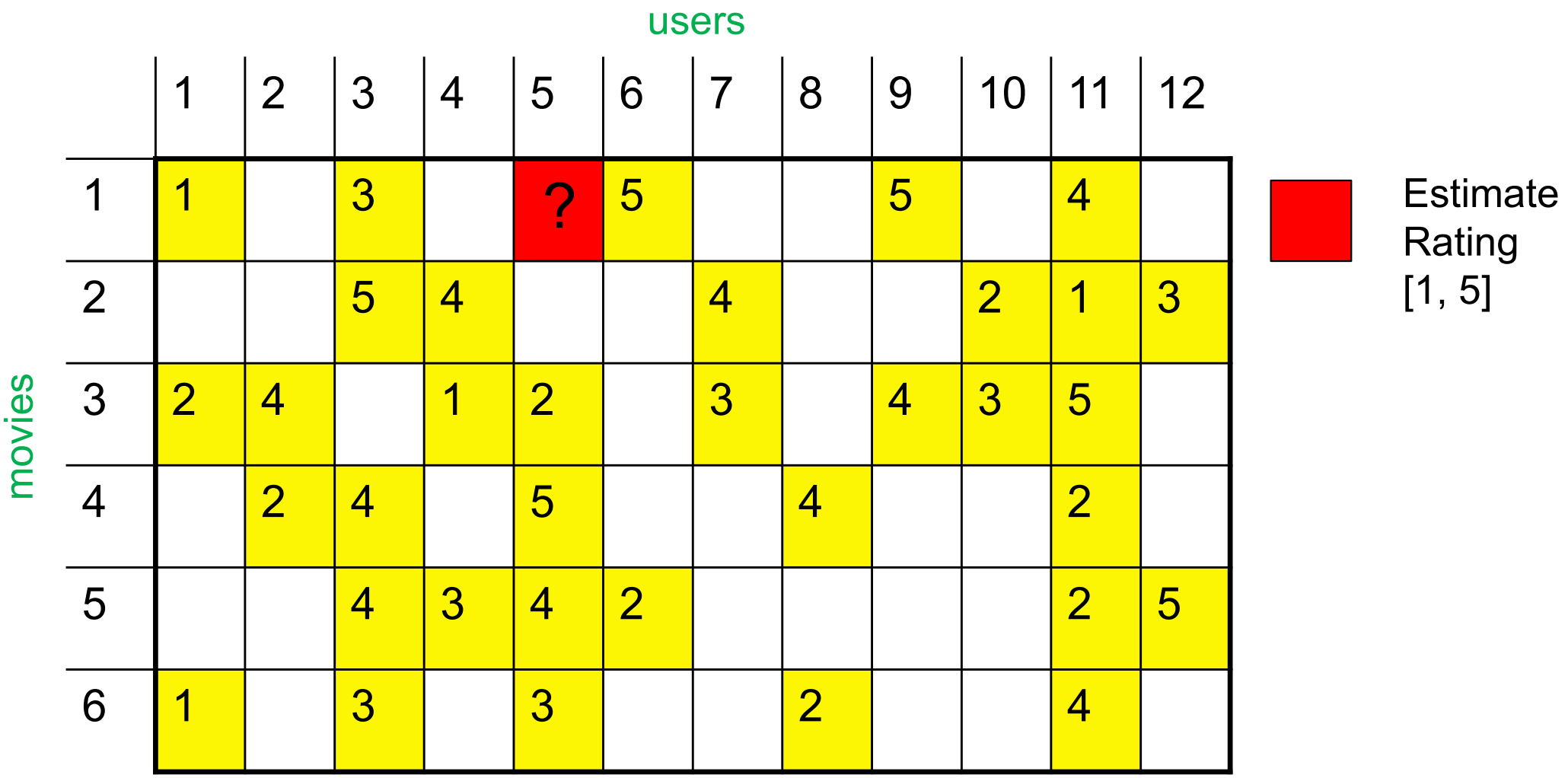
- **Item-Item CF (k = 2)**



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Collaborative Filtering

- **Item-Item CF (k = 2)**



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Collaborative Filtering

- **Item-Item CF (k = 2)**

movies \ users	1	2	3	4	5	6	7	8	9	10	11	12
1	1		3		?	5			5		4	
2			5	4			4			2	1	3
3	2	4		1	2		3		4	3	5	
4		2	4		5			4			2	
5			4	3	4	2					2	5
6	1		3		3			2			4	

PCC as a similarity measure:

1.) Subtract row mean m_i from each row

$$m_1 = (1 + 3 + 5 + 5 + 4) / 5 = 3.6$$

$$row_1 = [-2.6, 0, -0.6, 0, 0, 1.4, 0, 0, 1.4]$$

2.) Compute Cosine similarity between rows

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Collaborative Filtering

- **Item-Item CF (k = 2)**

users

movies	1	2	3	4	5	6	7	8	9	10	11	12	Sim(1, m)
1	1		3		?	5			5		4		1.00
2			5	4			4			2	1	3	-0.18
3	2	4		1	2		3		4	3	5		**0.41**
4		2	4		5			4			2		-0.10
5			4	3	4	2					2	5	-0.31
6	1		3		3			2			4		**0.59**

PCC as a similarity measure:

1.) Subtract row mean m_i from each row

$$m_1 = (1 + 3 + 5 + 5 + 4) / 5 = 3.6$$

$$row_1 = [-2.6, 0. -0.6, 0. 0. 1.4, 0, 0, 1.4]$$

2.) Compute Cosine similarity between rows

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Collaborative Filtering

- **Item-Item CF (k = 2)**

users

movies	1	2	3	4	5	6	7	8	9	10	11	12	Sim(1, m)
1	1		3		?	5			5		4		1.00
2			5	4			4			2	1	3	-0.18
3	2	4		1	2		3		4	3	5		**0.41**
4		2	4		5			4			2		-0.10
5			4	3	4	2					2	5	-0.31
6	1		3		3			2			4		**0.59**

Predict missing value by using weighted average:

r1,5 = (0.41 * 2 + 0.59 * 3) / (0.41 + 0.59) = 2.6

$$r_{xi} = \frac{\sum_{y \in N} sim_{xy} \cdot r_{yi}}{\sum_{y \in N} sim_{xy}}$$

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Collaborative Filtering

- **Item-Item vs. User-User**

	Avatar	LOTR	Matrix	Pirates
Alice	1		0.2	
Bob		0.5		0.3
Carol	0.2		1	
David				0.4

  - In practice: Item-Item works better!
  - Items are simpler, users have very different preferences

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Collaborative Filtering

- **Hybrid Methods**
  - Implement two or more different recommender systems
    - Estimate predictions by incorporating output from more recommenders
    - For instance, compute the final prediction as a linear combination
    - $P = (1 - \lambda) \cdot UU + \lambda \cdot II, \lambda \in [0, 1]$
  - Add content based approach features to CF
    - Introduce item profiles to address new item issue
    - Include demographic features to deal with new user issue

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Collaborative Filtering

- **Pros**
  - Works for various types of items
    - Universal approach, domain-specific knowledge not needed
- **Cons**
  - Cold Start
    - It needs users already in the system to provide predictions
  - Sparsity
    - Utility matrix is usually very sparse
    - Hard to find users that have rated same items
  - First Rater
    - It cannot recommend unrated item
    - New items, unpopular items
  - Popularity Bias
    - Black sheep problem (users with unique taste)
    - Favors popular items

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Remarks, Practice & Advices

- **Evaluation**

users

movies

1	2		3			3	
		3		3			
				2		1	
		3					
	2	4			5		2
				4	2	4	
		4					1

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Remarks, Practice & Advices

- **Evaluation**

users

movies

1	2		3			3	
		3		3			
				2		1	
		3					
	2	4			?		?
				?	?	?	
		4					?

Testing Data Set

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Remarks, Practice & Advices

- **Evaluating Predictions**
  - Compare predictions with known ratings
    - Root Mean Square Error (RMSE)
      - $\sqrt{\frac{1}{N}\sum_{x_i}\left(\widehat{r_{x_i}} - r_{x_i}\right)^2}$, where r_{x_i} is predicted and $\widehat{r_{x_i}}$ is the actual rating
    - Precision at top 10
      - % of those in top 10
    - Rank Correlation
      - Spearman's correlation between system's and user's rankings

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Remarks, Practice & Advices

- **Issues with Error Measures**
  - Focusing on accuracy may miss the point
    - Prediction Diversity
    - Prediction Context
    - Order of predictions
  - In recommenders, good performance for high ratings is important
    - RMSE emphases errors, method that performs well for high ratings and bad for other might get penalized

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Remarks, Practice & Advices

- **CF: Computational Complexity**
  - Expensive step: finding top k most similar users
    - $o(N \cdot M)$, N is a number of users, M is a number of items
  - Too expensive in runtime, solution is to precompute
    - For instance for User-User CF
    - $o(N^2 \cdot M)$
    - Obviously, it will not scale…
  - There is a better solution
    - Large scale neighbor identification
      - LSH, SimHash, MinHash
    - Clustering – reduces the space of potential solutions

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Remarks, Practice & Advices

- **Tip: Add more data**
  - All the data should be utilized
    - No use in data reduction to make fancy algorithms work
    - Simple methods on large data sets do best
  - Add even more data
    - E.g. for movies add IMDB data about genres
  - More data outperforms fancy algorithms

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State-of-the-art Approaches

- **Deep Learning / Neural CF**
  - Neural Collaborative Filtering
  - https://arxiv.org/abs/1708.05031
- **Transformer / Sequential Models**
  - BERT4Rec
  - https://arxiv.org/abs/1904.06690
- **Graph Neural Networks (GNN-based CF)**
  - LightGCN
  - https://arxiv.org/abs/2002.02126

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State-of-the-art Approaches

- **Self-Supervised / Contrastive Learning**
  - SGL
  - https://arxiv.org/abs/2010.10783
- **LLM / Generative Direction**
  - P5 (LLM-style recommender)
  - https://arxiv.org/abs/2203.13366

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